

GATE

ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 06



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Recap of Previous Lecture

Topic

Profit & Loss

Percent
(Terms)



Topics to be Covered



Topic-1

More on Profit Loss *el*

Topic-2

Mixtures & Alligation

[MCQ]



#Q. If the side of a square is increased by 20%, then what is the percentage change in its area?

Assignment

$$\begin{aligned} A &= \text{Side} \times \text{side} \\ &1.2 \times 1.2 \\ &= \underline{\underline{1.44}} \end{aligned}$$

A

44%

B

80%

C

22%

D

144%

2097 $\Rightarrow$ 2



Sold days

28th Nov, 20 98

FRIDAY

2000 $\Rightarrow$ 0

J $\rightarrow$ ~~3~~ J $\rightarrow$ ~~3~~

F $\rightarrow$ 0 A $\rightarrow$ ~~3~~

M $\rightarrow$ ~~3~~ S $\rightarrow$ ~~3~~

A $\rightarrow$ 2 O $\rightarrow$ 3

M $\rightarrow$ ~~3~~ N $\rightarrow$ 0

J $\rightarrow$ ~~2~~

3

97 $\Rightarrow$ 2

L4 N4

24 73

48

73

$$6 + 3 = 9 (2)$$

04
x24
96

[MCQ]



#Q. The population of a town doubled every 5 years from 2000 to 2015. What is the percentage increase in population in this period?

Assignment

A 800%

B 400%

C 700%

D 600%

$$2 \times 2 \times 2$$

$$= 8$$

$$700\%$$

$$700\% \uparrow$$

$$\rightarrow 2000 - 2005 \rightarrow 1$$

$$\rightarrow 2005 - 2010 \rightarrow 1$$

$$\rightarrow 2010 - 2015 \rightarrow 1$$

$$\frac{3}{3}$$

[MCQ]



#Q. If P is 60% taller than Q, by what percent is Q shorter than P?

Assignment

$$\frac{160}{100} \rightarrow \frac{100}{160} \times \frac{5}{8} = \underline{0.625}$$

A

40%

B

37.5%

C

62.5%

D

None of these

37.5%

0.375

[MCQ]



#Q. A is twice B and B is 200% more than C. By what percent is A more than C?

Assignment

$$A = 2B$$

$$B = 3C$$

$$A = 2 \times 3C$$

$$A = 6C$$

500% ↑

A 200%

B 400%

C 500%

D 600%

[MCQ]



#Q. The population of a village is 5500. If the number of males increases by 11% and the number of females increases by 20%, then the population becomes 6330. The population of the female in the village is?

- A 2000
- B 2500**
- C 3000
- D 3500

1.11 $\Rightarrow M + F = 5500$
 $1.11M + 1.2F = 6330$
 $1.11M + 1.11F = 6105$

 $0.09F = 225$

Assignment

$\therefore F = \frac{225}{0.09}$
 $= 2500$

[MCQ]



#Q. Rohan spends 40% of his monthly income on food items and 50% of the remaining on clothes and conveyance. He saves one-third of the remaining amount after spending on food, clothes and conveyance. If he saves Rs. 19200 every year, what is his monthly income?

- A 32000
- B 16000**
- C 12000
- D 6000

$1600 \leftarrow \text{Savvy per month} = \frac{19200}{12}$

$F \rightarrow 40\%$

$\text{Cloth \& Con} \rightarrow 30\%$ (60%)

$\text{Savvy} \rightarrow 10\%$ (30%)

$\frac{1}{3} \times 0.5 \times 0.6 = \frac{1}{3} \times 0.3 = 0.1$

$10\% \text{ of T.I.} = 1600$

$\text{T.I.} = \frac{1600}{0.1}$

$= 16,000$

[MCQ]



#Q. 5% of income of P is equal to 15% of income of Q and 10% of income of Q equal 20% of income of R. If R's income is 2000, then What is total income of P, Q and R?

$$5\% \text{ of } P = 15\% \text{ of } Q$$

Assignment

$$P = 3Q$$

$$10Q = 20R \Rightarrow Q = 2R$$

$$R = 2000$$

$$Q = 4000$$

$$P = 12000$$

$$18k$$

A 9000

B 12000

C 15000

D 18000

PROFIT & LOSS

Investment

C.P.

P
C.P. < S.P.

gain / profit

L
C.P. > S.P.

g/l / P/l

No P/L
C.P. = S.P.

Return

S.P.

No Disc

1000 x

21.

~~Marked Price~~
Marked price

1700

discount

Labelled 21.

List

Tagged

PROFIT & LOSS

$$\underline{S.P. > C.P.}$$

= Profit ✓

$$\underline{S.P. < C.P.}$$

= Loss ✓

$$\underline{S.P. = C.P.}$$

= No Profit No Loss ✓

$$\frac{S.P.}{C.P.} > 1$$

$$\frac{S.P.}{C.P.} < 1$$

$$\frac{S.P.}{C.P.} = 1$$

S.P.

C.P.

PROFIT & LOSS

Note:



A handwritten diagram inside a circle. It shows 'S.P.' (Selling Price) above a horizontal line, and 'C.P.' (Cost Price) below the line. This represents the relationship between selling price and cost price.

Profit or loss percentage is to be applied always to the Cost Price only.

Discount percentage is to be applied always to the Marked Price only.

[MCQ]



#Q. If selling price and cost price are in the ratio 8:5, then find the profit% or loss%.

$$\underline{1.6} = \frac{8}{5} = \frac{S.P}{C.P}$$

6

60% Profit

[MCQ]



#Q. A Fruit seller purchases 11 orange for Rs. 10 and sells 10 orange for Rs. 11.

If he follows the same process, then, find his profit or loss%?

$$C.P \rightarrow \frac{10}{11}$$

$$S.P = \frac{11}{10}$$

$$\frac{S.P}{C.P} = \frac{11}{10} \div \frac{10}{11}$$

$$= \frac{11}{10} \times \frac{11}{10} =$$

$$\frac{121}{100} = 1.21$$

$$10\% P$$

$$10\% P$$

$$1.1 \times 1.1 = 1.21$$

$$21\% P$$

$$\frac{S.P.}{C.P.}$$

[MCQ]



#Q. A milk vendor purchases milk at Rs. 72/ litre, and sells at Rs. 60/ litre. For every 1 litre milk he adds 200ml. of water. While selling milk he cheats 200ml. in 1 liter measurement. Find his Profit or Loss percentage.

25% Profit

$$\frac{S.P}{C.P} = \frac{60}{72} \times \frac{1200}{1800} \times \frac{1800}{800}$$
$$= \frac{5}{4} = 1.25$$

[MCQ]

Chain rule ✓



#Q. A milk vendor purchases milk at Rs. 72/ litre, and sells at Rs. 60/ litre. For every 1 litre milk he adds 200ml. of water. While selling milk he cheats 200ml. in 1 liter measurement. Find his Profit or Loss percentage.



[MCQ]



#Q. A cloth merchant purchases cloth at ₹80/meter and sells at ₹100/meter. As a festive offer, he gives 50% extra free on every demand. Find his profit% or loss%.

A

25% profit

B

8.33% profit

C

16.66% loss

D

83.33% profit

Handwritten calculation:

$$\frac{S.P.}{C.P.} = \frac{100}{160} \times \frac{1}{1.5} = \frac{100}{240} = \frac{5}{12}$$

Handwritten calculation:

$$\frac{2}{3} \text{ OR } \frac{4}{6} \text{ OR } \frac{100}{150}$$

Handwritten calculation:

$$= \frac{5}{6} = 0.8\bar{3}$$

[MCQ]



#Q. Due to downfall in the market, 'A' sells mangoes to 'B' saying "FOR EVERY 12 MANGOES, COUNT AS 8". Due to overnight demand, 'A' took his mangoes back from 'B' saying "FOR EVERY 8 MANGOES, COUNT AS 12". Find the profit or loss percent of 'A'.

$$\underline{55.5\% \text{ loss}}$$

$$\frac{28}{12} \times \frac{8}{12}$$

$$= \frac{4}{9} = 0.\overline{4} = 0.\overline{5}$$

PROFIT & LOSS

Note:

A trader may sometimes have multiple profits or losses simultaneously.

This is equivalent to having multiple changes and so all individual changes are to be multiplied to get the overall effect.

[MCQ]



#Q. Sunidhi bought 15 apples for Rs.10 and sold them at the rate of 12 apples for Rs.12. What is the percentage of profit made by her?

$$C.P = \frac{10}{15} \quad | \quad S.P = \frac{12}{12} = 1$$

A 100%

B 150%

C 125%

D None of these

$$\frac{S.P}{C.P} = 1 \times \frac{15}{10} = 1.5$$

50% Profit

[MCQ]



#Q. A shopkeeper advertises for selling cloth at 4% loss. However by using a false meter scale he actually gains 25%. What is actual length of scale?

- A** 76.8 cm
- B** 77.8 cm
- C** 74.8 cm
- D** 75.8 cm

$$\frac{S.P.}{C.P.} = 0.96 \times \frac{100}{x} = 1.25$$
$$\Rightarrow \frac{96}{x} = 1.25$$
$$\Rightarrow \frac{96}{1.25} = x$$
$$\therefore x = \frac{9600}{125} = 76.8 \text{ cm}$$

[MCQ]



#Q. A man sells an article at a profit of 20%. If he had bought it at 10% less and sold it for Rs. 18 more, he would have gained 40%. Find the cost price of the article.

Assignment

[MCQ]



#Q. An article was sold at a profit of 20%. If both cost price and selling price are ₹100 less each, then magnitude of the percentage of profit would have been 4 percentage points more than that in the first case. Then the cost price is

Assignment

- A** ₹ 500
- B** ₹ 600
- C** ₹ 800
- D** None of these

[MCQ]



#Q. 5kg of ghee was bought by Vinod for ₹300. One kg from spoilt. He sells the remaining in such a way that on the whole he incurs a loss of 10%. At what price per kg does he sell the ghee?

Assignment

- A** ₹ 46.25
- B** ₹ 45.70
- C** ₹ 67.50
- D** ₹ 46.60

Puzzle:

Assignment



- #Q. A function hall was filled with 100 guests including men, women and kids. 100 biscuits has to be distributed among these guests, such that each man gets 4 biscuits, each woman as 3 and each kid gets $\frac{1}{2}$ biscuits. How many men, women and kids are there in function hall?



2 mins Summary



Topic

~~Mixtures & Alligations~~

More on Profit & Loss

Discount? 4

Mix & Allig

THANK - YOU